

Modern storage systems typically support rich data services like data replication, snapshot management, file system level encryption and compression services. While such rich data services cater to today's demand for storage, the brave new world of IoT requires a revisit of the promise of storage appliances from dumb data containers to actionable insight generators. Raw data merely does not provide actionable value for end customers, so delivering insight-centric storage should be the key direction along which future storage systems need to evolve.

There has been lots of interest around active data services that help to mine data and offer data discovery. 'Deep storage' contains historical data. 'Change data capture' (CDC) in the context of the existing *data at rest*, allows us to answer the following questions:

- Does the new incoming piece of data, when combined with a summary/impression of existing long tailed deep storage data, give rise to new insights?
- Does it allow us to invalidate any existing insights that we have derived in the past?
- Is the new incoming piece of data interesting enough (anomaly/outlier) that it needs to be kept sight of or can it be discarded?

Here are some examples:

- For a content serving network, incoming location data about a client's movement from say, Karnataka to Gujarat, triggers a data replication event to host the client's favourite Kannada film songs/movies onto an edge network which is faster to access in Gujarat.
- For an advertising company's client website, an update in the email ID of long-standing clients indicates their move to being a top corporate customer. This is an insight of loyal supporters that your sales team will now have in your top customers' accounts.
- Energy meter data, coming in daily from an IoT sensor for a group of households served by a common solar panel, suddenly shooting up may indicate a malfunction in the common service infrastructure and will allow the solar company to initiate proactive maintenance.

From the above use-cases, it is clear that any new incoming piece of data needs to be seen in the context of existing historical data in order to deliver insights. Incoming data also needs to be seen in the context of the consumer/client interested in the data, from whom the insight is significant. Privacy considerations may need anonymous insights applicable to a target population/group of users. Relevance and significance of incoming data is best noted at the point of ingest in the frame of existing historical data and current context.

Just consider this scenario. A Client X has a set of data stores associated with him. One data store is his entertainment album, one is his work folder containing his software projects and one is his financial records. Your intelligent storage system has tagged these folders based

on the access time and historical patterns, inferring that your client accesses the entertainment data store between 7 p.m. and 10 p.m. Given that your tenant has purchased storage, you would like to improve the quality of service given to him. Knowing his usage pattern, you can proactively move his media album to low latency Flash storage automatically between 7 p.m. and 10 p.m., thereby improving his entertainment experience.

The arrival of the Internet of Things (IoT) heralds a new way of looking at how storage has traditionally been used. Smart IoT devices typically come with a small amount of storage (capable of storing one or two days of data, at the most) and, over longer periods, push data to cloud-backed stores for long tail analysis. Local storage at smart IoT devices also needs to cater to operating conditions of low power consumption and intermittent connectivity. Hence, analytic algorithms need to be power-aware and resilient even in the face of intermittent network connectivity.

Increasingly, smart IoT devices will create digital footprints in a physical world. The data that is generated locally in the context of a physical entity (like a user, machine or software application) gets pushed to separate silos in the cloud and is analysed in isolation, without factoring in the context in which a particular physical entity generated these multiple data sources. While data in each of these individual silos get analysed and mined, insight discovery across these different data sources is typically not achieved due to storage silos. In the context of the IoT, the digital footprints get lost when moved to separate silos. However, at the common origin of the generating entity, they can be unified and analysed for insight discovery in context. What is needed is local analytics at the origin of the data generating entity, with local context and realtime responses. Hence, there needs to be a shift in the current IoT data analytics paradigm. While the current paradigm is to store the raw data locally, then move to the cloud, before analysing it and acting on the insights, it makes sense to do local filtering and analysis, gain insights applicable locally and only then store, if needed, in the cloud. If not required, the raw data can be discarded.

There has been a lot of debate over deductive and predictive analytics and their ability to mine for insights and patterns. While computers have been extremely good in learning certain kinds of problems, such as finding the best route to a destination or enumerating all combinations, they rarely beat humans in identifying patterns, especially when the features are approximate. A typical example is the ability of human intuition to tag a face as that of a friend or foe within a time that is 100 times faster than any machine.

There are various kinds of data discovery problems that are encountered. Certain problems require precise prediction with information on how the prediction was made. For instance, consider the selection of a candidate for a job.